ALGEBRA 1 · UNIT 5: LINEAR FUNCTIONS AND GRAPHING

Balloon Festival

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

At the Sky Valley Balloon Festival, each balloon's altitude is given a different way: a table, an equation, a graph, and words. Students find rates of change and starting altitudes, write equations, predict when balloons meet or land, and design a balloon that meets festival rules.

Students practice moving between the four representations of a linear function and use slope and intercept to compare situations. Day 2 uses equations to predict events (landing, catching up). Day 3 asks students to design a balloon that satisfies constraints.

Standards Addressed

- F-IF.6 Interpret the rate of change
- F-IF.9 Compare functions represented in different ways
- F-LE.2 Construct linear functions from a graph, a description, or a table
- A-CED.2 Create equations in two variables

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with the balloon picture: Which balloon is climbing fastest? Students can't tell yet! Part 1 rates and starting altitudes.
Day 2	Part 2: write equations and solve to predict events. Graph all balloons on one grid.
Day 3	Part 3: design a balloon; create the flight report.

Materials

- Student handout
- Graph paper or the grid provided
- Colored pencils (one color per balloon)
- Calculators

How the Three Versions Differ

Level	What changes
CP2	Rates and intercepts are found with prompts; equations are evaluated and solved for one balloon at a time.
CP1	Adds a balloon given by two points and asks when balloons reach the same altitude (variables on both sides).
Honors	Adds standard form, unit conversion, a general meeting-time formula, a drift model with x and y, and a design whose slope must satisfy an inequality.

Watch For (Common Misconceptions)

- Using the time step (2 minutes) as the rate instead of dividing the change in altitude by the change in time.
- Reading the y-intercept from the wrong axis on the graph.
- Thinking a negative rate means the balloon is below the ground.

Extension Ideas

- Add a balloon described by a story with a pause (piecewise)
- Research the real ascent rates of hot air balloons.
- Animate the balloons in Desmos using the equations.

Answer Key – CP2

Part 1: Read the Judges' Data

#	Task	Answer
1	Balloon A (table): Find the rate of change and the starting altitude.	Rate = 120 ÷ 2 = 60 ft per minute; starts at 150 ft.
2	Balloon B (equation): y = 45t + 300. What is the rate of change? What is the starting altitude?	Rate 45 ft per minute; starts at 300 ft.
3	Balloon C (graph): Find the starting altitude and the rate of change. Is the balloon going up or down?	Starts at 900 ft; rate = (400 − 900) ÷ 10 = −50 ft per minute; going down.
4	Balloon D (words): "Balloon D launched from a hill 75 feet above the field and rises 80 feet every minute." Find its rate and starting altitude.	Rate 80 ft per minute; starts at 75 ft.
5	List the balloons from fastest rising to slowest. Which balloon is coming down?	D (80), A (60), B (45). C is coming down (−50).

Part 2: Equations & Predictions

#	Task	Answer
6	Write equations for Balloons A, C, and D.	A: y = 60t + 150. C: y = −50t + 900. D: y = 80t + 75.
7	Find the altitude of each balloon after 5 minutes. Which is highest?	A 450 ft, B 525 ft, C 650 ft, D 475 ft. C is highest.
8	When will Balloon A reach 1,050 feet? Write and solve an equation.	60t + 150 = 1050 → 60t = 900 → t = 15 minutes
9	When will Balloon C land (altitude 0)?	−50t + 900 = 0 → t = 18 minutes

Part 3: Enter Your Own Balloon

#	Task	Answer
10	Choose a starting altitude and a rate of change for your balloon. Write its equation.	Answers vary.
11	Make a table for t = 0, 2, 4, 6, 8, and 10 minutes and graph your balloon with Balloons A and B.	Answers vary.
12	Is your balloon higher or lower than Balloon A at 10 minutes? How do you know?	Answers vary; A is at 750 ft at t = 10.

Answer Key – CP1

Part 1: Read the Judges' Data

#	Task	Answer
1	Balloon A is given in the table.	$y = 60t + 150$
2	Balloon C is given in the graph.	$y = -50t + 900$
3	Balloon D: "Launched from a hill 75 feet above the field and rises 80 feet per minute."	$y = 80t + 75$
4	A fifth balloon, E, was spotted at 420 feet at 3 minutes and at 700 feet at 7 minutes. Write its equation and interpret its y-intercept.	$m = (700 - 420) \div (7 - 3) = 70$; $420 = 70(3) + b \rightarrow b = 210$; $y = 70t + 210$. It started 210 ft high.
5	Balloon B is $y = 45t + 300$. Compare B and E: which started higher, and which is rising faster?	B started higher (300 vs. 210); E is rising faster (70 vs. 45).

Part 2: When Do They Meet?

#	Task	Answer
6	When are Balloons A and B at the same altitude? What is that altitude?	$60t + 150 = 45t + 300 \rightarrow 15t = 150 \rightarrow t = 10$ min, 750 ft
7	When does Balloon D pass Balloon B? Give the time in minutes and seconds.	$80t + 75 = 45t + 300 \rightarrow 35t = 225 \rightarrow t \approx 6.43$ min ≈ 6 min 26 s
8	Balloon A is going up and Balloon C is coming down. When are they at the same altitude?	$60t + 150 = -50t + 900 \rightarrow 110t = 750 \rightarrow t \approx 6.82$ min (about 559 ft)
9	The festival ceiling is 1,500 feet. Which rising balloon reaches the ceiling first? Show your work.	A: 22.5 min; B: 26.67 min; D: 17.81 min; E: 18.43 min. D reaches it first.

Part 3: Enter Your Own Balloon

#	Task	Answer
10	Find when Balloon B reaches 1,000 feet.	$45t + 300 = 1000 \rightarrow t \approx 15.56$ minutes
11	Design your balloon. Write its equation and show it meets both rules.	Answers vary; check $100 \le b \le 300$ and $(1000 - b) \div m < 15.56$.
12	Graph Balloons A, B, and yours on one grid and mark every meeting point.	Answers vary.

Answer Key – Honors

Part 1: Read the Judges' Data

#	Task	Answer
1	Balloons A (table), C (graph), and D ("launched from a hill 75 feet above the field, rising 80 feet per minute").	A: $y = 60t + 150$. C: $y = -50t + 900$. D: $y = 80t + 75$.
2	Balloon E was at 420 feet at 3 minutes and at 700 feet at 7 minutes.	$y = 70t + 210$
3	Balloon F is given in standard form: $150t - 3y = -600$. Rewrite it in slope-intercept form and interpret.	$-3y = -150t - 600 \rightarrow y = 50t + 200$. Starts at 200 ft, rises 50 ft per minute.
4	Convert Balloon A's rate to feet per second. Write A's equation with time s in seconds.	60 ft per min = 1 ft per second; $y = s + 150$

Part 2: General Meeting Times

#	Task	Answer
5	Derive a formula for the time t when they have the same altitude. When does the formula fail, and what does that mean?	$m_1t + b_1 = m_2t + b_2 \rightarrow t = (b_2 - b_1) \div (m_1 - m_2)$. It fails when $m_1 = m_2$ (parallel): they never meet (or are always together if $b_1 = b_2$).
6	Use your formula to find when A meets B, and when D meets B.	A and B: $(300 - 150) \div (60 - 45) = 10$ min. D and B: $(300 - 75) \div (80 - 45) \approx 6.43$ min.
7	Do Balloons C and F ever meet? Do Balloons B and a new balloon $y = 45t + 200$ ever meet? Explain using slopes.	C and F: $t = (200 - 900) \div (-50 - 50) = 7$ min, at 550 ft, so yes. B and $y = 45t + 200$: same slope, different start, so never.
8	Wind pushes Balloon A east 300 feet per minute. Write the horizontal distance x as a function of t, then write A's altitude y as a function of x....	$x = 300t \rightarrow t = x \div 300$. $y = 60(x \div 300) + 150 = 0.2x + 150$. The balloon rises 0.2 ft for every foot it drifts east.

Part 3: Design Within the Rules

#	Task	Answer
9	Write the rate m in terms of the starting altitude b and the time T when the balloon reaches 1,000 feet.	$m = (1000 - b) \div T$
10	Find the smallest and largest possible rates that follow the rule. Write the result as a compound inequality.	Smallest: $(1000 - 200) \div 10 = 80$. Largest: $(1000 - 100) \div 8 = 112.5$. So $80 \leq m \leq 112.5$.

11	Choose a design, show it meets every rule, and graph it with A, B, and D. Will your balloon meet each of them? When?	Answers vary.

Scoring Rubric (all levels)

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Your balloon meets the festival rules, with an equation, table, and graph that match. Goes beyond the minimum with a creative, realistic choice.	Your balloon meets the festival rules, with an equation, table, and graph that match.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Flight report shows all balloons as equations and graphs, answers the judges' questions, and presents your balloon. Polished, organized, and visually engaging.	Flight report shows all balloons as equations and graphs, answers the judges' questions, and presents your balloon. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.